



Assignment #02 (Lab)

Course: Information Security (CSC232)

Class: BCS – 4B

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Marks: 50

Submission Deadline: 12-Oct-2025

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Q1. AES with Classical Pre-Processing

Before encrypting with AES, first apply a Caesar cipher shift of 3 to the plaintext.

Step 1: Take user input text.

Step 2: Apply Caesar cipher.

Step 3: Encrypt the Caesar output using AES in CBC mode.

Step 4: Decrypt and verify correctness.

Q2. DES Double Encryption

- Implement Double DES (plaintext $\rightarrow$ DES(K1) $\rightarrow$ DES(K2)):
- Write functions for encryption and decryption.
- Test with two different keys.
- Show what happens if $K1 = K2$ (does it behave like normal DES?).

Q3. AES vs DES Block Size Analysis

- Encrypt the same plaintext message with:
- AES in ECB mode
- DES in ECB mode

Then:

- Print ciphertexts in hex.
- Compare block sizes (AES = 128 bits, DES = 64 bits).
- Write 4–5 lines on why AES is more secure than DES apart from key size.

Q4. File Fragment Encryption with AES and DES

Write a program to:

- Take a text file and split it into two halves.
- Encrypt the first half with AES (CBC) and the second half with DES (ECB).
- Combine both ciphertexts and save them to a new file.
- Write a decryption routine that correctly recovers the original file.

Q5. Bit-Flipping Experiment

- Encrypt a short message with:
- AES-CBC
- DES-CBC

Then flip 1 bit in the ciphertext and decrypt.

- Compare how much the plaintext changes in AES vs DES.
- Explain (in 4–5 lines) why error propagation differs between block sizes and modes.